F78AA - Actuarial and Financial Mathematics A

Tutorial 6 – Bonds and Tax

Question 1 A government has just issued a fixed interest security (or bond) of nominal amount £100 pays a coupon of 6% p.a. at the end of each of the next ten years and is redeemed at par at the end of the ten years.

1. Determine the price of the security which gives a redemption yield of 6% p.a. effective to a purchaser who pays no tax. Explain the answer.
2. Calculate the price of the security that gives a redemption yield to the same purchaser of 8% p.a. effective.

Question 2 Suppose that in the above example the coupon is £6 p.a. payable half-yearly.

1. Calculate the price of the security that gives a yield to the same purchaser of: (a) 6% p.a. effective, and (b) 8% p.a. effective.
2. Explain why your result in (a) is not £100.

Question 3 A fixed interest security of nominal amount £100 000 will be repaid at 110% after 15 years. The security bears a coupon of 9% p.a. payable annually in arrears. On the issue date, the security is purchased for £80 000 by an investor who is liable to income tax at 40% and to capital gains tax at 30%.

1. Calculate the investor's net (meaning "after tax") effective redemption yield p.a. for the transaction.
2. The investor wishes to secure an after-tax redemption yield of 9% p.a. Determine how much the investor should pay for the security.

Question 4 Consider a 20-year bond, bearing a half-yearly coupon of 8% per annum and redeemable at 120%. It is exactly six months until the next coupon. Calculate the price of the bond to give a redemption yield of 9% per annum effective after tax to an investor who pays:

1. no taxes.
2. income tax at 20% but no capital gains tax (CGT).
3. income tax at 20% and CGT at 30%.

Question 5 On 1 January 2003, a corporation issues a number of bonds, each of nominal amount £100, to be redeemed at 98% on 1 January 2028. The coupon is 7% per annum, payable half-yearly on each 1 January and 1 July.

1. The bond is priced at issue to give a redemption yield of 8.16% per annum to a non-tax-paying investor. Calculate the issue price per £100 nominal.
2. Anna, who pays income tax at 25% and CGT at 40%, bought some of the bonds on the issue date. On 1 July 2013, just after the coupon then due was paid, she sold the bonds to Bob, who pays both income tax and CGT at 40%. Bob paid a price that gave him a redemption yield after taxes of 6% p.a. effective. Calculate the price that Bob paid per £100 nominal.
3. Calculate the yield realised by Anna on her completed transaction.

Question 6 Investors often allow for risk by requiring a higher yield to lend money to a borrower who is thought to be at higher risk of not repaying some or all of the interest and capital. Explain the consequences for companies or governments who are trying to raise money in the bond markets.

Question 7 An investor buys a 20-year bond, to be redeemed at 110% and bearing semi-annual coupons at rate 6% per annum, to obtain a redemption yield of 5% p.a. effective. The investor pays income tax at 25% but no CGT.

1. Calculate the price paid by the investor.

2. Just after the 20th coupon payment, the investor's rate of income tax drops to 20%. If she holds the bond to redemption, calculate her realised yield on the whole transaction.

Question 8 A fixed interest stock with nominal value of £100 was issued on 1 January 1995. The coupon rate was 4% p.a. payable half-yearly in arrears for the first 10 years and then 3% p.a. payable half-yearly in arrears for the next 10 years. Redemption was at par after 20 years. On 1 July 1999, just after the July dividend was paid, the stock was quoted at a price which would have given a tax-free investor a redemption yield of 6% p.a. The stock was bought on that date by an investor who was subject to income tax at 40% and to capital gains tax at 20%.

1. The investor held the bond until the redemption date. Calculate his after tax redemption yield.

2. On 1 August 2000 the stock was priced to give an after tax redemption yield of 5% p.a. effective to the taxed investor described above. Calculate the price of the stock on 1 August 2000. (Note: be careful with the dates.)

Solutions: Q1.1. £100; Q2.1. £86.57984; Q2.1(a). £100.65278; Q2.1(b). £87.37017; Q3.1. 0.0772; Q3.2. £70.47126; Q4.1. £96.04715; Q4.2. £81.12001; Q4.3. £78.92111; Q5.1. £88.97748; Q5.2. £79.42754; Q5.3. 5.12% p.a.; Q7.1. £98.23044; Q7.2. 5.001% p.a.; Q8.1. 3.999% p.a.; Q8.2. £67.17545.

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Solutions to Tutorial 6 – Bonds and Tax

Solution 1

1. The price is £100. This is because it satisfies the Principle of Equivalence:

$$100 = 6\, a_{\overline{10}|}^{0.06} + 100 \times 1.06^{-10}.$$

2. If the desired yield is 8% p.a. effective, we must solve:

$$P = 6\, a_{\overline{10}|}^{0.08} + 100 \times 1.08^{-10} \Rightarrow P = £86.57984.$$

Why would we not round the answer to, e.g. 2 d.p.s? Prices of bonds are usually expressed as the price of buying £100 nominal of the bond. However, investors would buy many more pounds nominal of the bond than £100. Usually, investors buy nominal amounts in excess of £10 000 nominal.

Solution 2

1. Working in time units of half-years, we solve

$$P = \frac{6}{2}\, a_{\overline{20}|}^{0.029563} + 100 \times 1.029563^{-20} \quad \Rightarrow \quad P = £100.65278.$$

This is greater than £100 (as in Q.1) because part of the annual interest payment of £6 p.a. is received slightly earlier, so the bond is worth more.

2. Working in time units of half-years, we solve

$$P = 3\, a_{\overline{20}|}^{0.039230} + 100 \times 1.039230^{-20} \quad \Rightarrow \quad P = £87.37017.$$

Solution 3

1. Capital Gains Tax (CGT) is payable, so the equation of value (per £100 nominal) is:

$$\begin{aligned} f(i) &:= -80\,(1+i)^{15} + 0.6 \times 9 \times s_{\overline{15}|}^{i} + 110 - 0.3(110-80) = 0 \\ &\Rightarrow -80\,(1+i)^{15} + 5.4\, s_{\overline{15}|}^{i} + 101 = 0. \end{aligned}$$

We use linear interpolation to find the solution $i^\star$ to the equation of value. By trial and error:

$$\begin{aligned} f(0.07) &= +15.974 \\ f(0.08) &= -6.152 \\ \Rightarrow i^\star &\approx 0.07 + 0.01\, \frac{15.974}{15.974 + 6.152} = 0.0772. \end{aligned}$$

2. Since $0.09 > 0.0772$ we know the price paid will be less than £80 000 so CGT will certainly be payable. Therefore by the Principle of Equivalence:

$$\begin{aligned} P &= 5.4 a_{\overline{15}|}^{0.09} + (110 - 0.3(110 - P))\, 1.09^{-15} \\ \Rightarrow P(1 - 0.3 \times 1.09^{-15}) &= 5.4 a_{\overline{15}|}^{0.09} + 77 \times 1.09^{-15} \\ \Rightarrow P &= £70.47126. \end{aligned}$$

Solution 4 In each case we switch to a time unit of a half-year, and a redemption yield of 0.044031 per half-year effective. Per £100 nominal:

1. $P = 4\, a_{\overline{40}|}^{0.044031} + 120 \times 1.044031^{-40} \Rightarrow P = £96.04715.$
2. $P = 3.2\, a_{\overline{40}|}^{0.044031} + 120 \times 1.044031^{-40} \Rightarrow P = £81.12001.$
3. $(1 - t_1)\, d/R = 0.02667 < 0.044031$ so CGT is payable.

$$\begin{aligned} P &= 3.2\, a_{\overline{40}|}^{0.044031} + (120 - 0.3(120 - P)) \times 1.044031^{-40} \\ \Rightarrow P(1 - 0.3 \times 1.044031^{-40}) &= 3.2\, a_{\overline{40}|}^{0.044031} + 84 \times 1.044031^{-40} \\ \Rightarrow P &= £78.92111. \end{aligned}$$

Solution 5

1. The term is 25 years. We use a time unit of half a year, and a yield of 0.04 per half-year effective.
$$P = 3.5\, a_{\overline{50}|}^{0.04} + 98 \times 1.04^{-50} \Rightarrow P = £88.97748.$$
2. On 1 July 2013 the bond has 14.5 years (29 half-years) to run. Bob's effective yield is 0.029563 per half-year. From Bob's tax position $(1 - t_1)\, d/R = 0.02143 < 0.029563$ so Bob has to pay CGT.
$$\begin{aligned} P &= 2.1\, a_{\overline{29}|}^{0.029563} + (98 - 0.4(98 - P)) \times 1.029563^{-29} \\ \Rightarrow P(1 - 0.4 \times 1.029563^{-29}) &= 2.1\, a_{\overline{29}|}^{0.029563} + 58.8 \times 1.029563^{-29} \\ \Rightarrow P &= £79.42754. \end{aligned}$$
Anna's financial and tax position are irrelevant to Bob's calculation.
3. Anna sold the bonds for less than the price paid for them so does not pay any CGT. She had held the bonds for 10.5 years. The equation of value, based on Anna's tax position and a time unit of half a year, is
$$f(i) := -88.97748 \times (1 + i)^{21} + 0.75 \times 3.5 \times s_{\overline{21}|}^{i} + 79.42754 = 0.$$
However, to make it easier to plug the equation of value into our calculator, we round the buying and selling prices to 2 d.p.s and set
$$f(i) := -88.98 \times (1 + i)^{21} + 0.75 \times 3.5 \times s_{\overline{21}|}^{i} + 79.43 = 0.$$
We use linear interpolation to find the solution $i^\star$ to the equation of value. By trial and error (we know that Anna's yield must be less than about $0.75 \times 3.5/88.98 \approx 3\%$ per half year):
$$\begin{aligned} f(0.02) &= +12.247 \\ f(0.03) &= -10.823 \\ \Rightarrow i^\star &\approx 0.02 + 0.01\, \frac{12.247}{12.247 + 10.823} = 0.0253. \end{aligned}$$
Hence Anna's effective annual yield is approximately 5.12%.

Solution 6 Yields are inversely related to price, so for a fixed set of future cashflows, the higher the yield, the lower the price today of the future cashflows. By demanding a higher yield from borrowers thought to be at greater risk of defaulting, lenders make it more expensive for high-risk borrowers to raise money (a company or government who issues a bond is borrowing money). From the lender's point of view they require a higher reward for taking on the risk of the issuing company defaulting on the interest and capital repayments, an example of the relationship between risk and reward that underlies most of financial mathematics.

Solution 7

1. Using a time unit of half a year and an effective yield of 0.024695 per half year:

$$P = 2.25\, a_{\overline{40}|}^{0.024695} + 110 \times 1.024695^{-40} \Rightarrow P = £98.23044.$$

2. The equation of value can be written as (where, as before, we round the price P to 2 d.p.s for ease of calculation)

$$f(i) := -98.23\,(1+i)^{40} + 2.25\, s_{\overline{20}|}^{i} \times (1+i)^{20} + 2.4 s_{\overline{20}|}^{i} + 110 = 0.$$

As the coupon payments are relatively small and the change in the tax rate is not too large, we do not expect a large change in the yield. We use linear interpolation to find the solution $i^\star$ to the equation of value. By trial and error, in time units of half a year:

$$\begin{aligned} f(0.02) &= -32.653 \\ f(0.03) &= +36.747 \\ \Rightarrow i^\star &\approx 0.02 + 0.01\,\frac{32.653}{32.653 + 36.747} = 0.0247. \end{aligned}$$

Hence the annual yield is approximately 5.001% effective. (Note: this answer would be acceptably accurate in an exam, however you should spot that it is unlikely that a change amounting to 5% of half the coupons paid should have such a small effect on the yield. Interpolation between $i = 0.025$ and $i = 0.026$ gives a more accurate yield of about 5.10% per annum effective.)

Solution 8

1. On 1 July 1999 the 9th coupon has just been paid, and the outstanding term is 15.5 years (31 half years). Using a time unit of half a year, the price on that date was:

$$P = 1.5\, a_{\overline{31}|}^{0.029563} + 0.5\, a_{\overline{11}|}^{0.029563} + 100 \times 1.029563^{-31} \Rightarrow P = £75.3413 \qquad (75.34 \text{ rounded}).$$

On redemption, CGT was payable. The CGT is $20\% \times (100 - P) = £4.93174$. After payment of the CGT from the redemption proceeds, the investor is left with £95.06826.

The equation of value, in half-year time units, can be written as:

$$f(i) = -75.34\,(1+i)^{31} + 1.2\, s_{\overline{31}|}^{i} - 0.3\, s_{\overline{20}|}^{i} + 95.07 = 0.$$

We know this investor must get a lower yield than the yield that the tax-free investor obtained. We use linear interpolation to find the solution $i^\star$ to the equation of value. By trial and error, in time units of half a year:

$$\begin{aligned} f(0.01) &= +29.261 \\ f(0.02) &= -0.561 \\ \Rightarrow i^\star &\approx 0.01 + 0.01\,\frac{29.261}{29.261 + 0.561} = 0.0198. \end{aligned}$$

Hence the annual yield is approximately 3.999% effective.

2. On 1 August 2000, the outstanding term is 14 years and 5 months, or 28.833 half-years, and 29 coupons remain to be paid. The test for CGT to be payable does not work exactly since the coupon is not level, but using the higher level of coupon payment, $(1-t_1)\,d/R = 0.012 < 0.024695$ so we may be confident that CGT is payable. Then the price is the price just after payment of the coupon paid on 1 July 2000, accumulated for 1 month, which can be written:

$$\begin{aligned} P &= 1.024695^{1/6}(0.9\,a^{i}_{\overline{29}|} + 0.3\,a^{i}_{\overline{9}|} + (100 - 0.2(100 - P))\,(1.024695)^{-29}) \\ &\Rightarrow \frac{P(1 - 0.2 \times (1.024695)^{-29})}{1.024695^{1/6}} = 0.9\,a^{i}_{\overline{29}|} + 0.3\,a^{i}_{\overline{9}|} + 80(1.024695)^{-29} \\ &\Rightarrow P = £67.17545. \end{aligned}$$